Lecture 5: Switching Techniques Computer Networks

Introduction to Switching

In a computer network, many devices communicate with one another. Connecting every device directly to every other device would require a large number of communication links and would be expensive to maintain.

Switching provides an efficient way to transfer data between devices through intermediate network nodes. These nodes receive data and direct it toward the appropriate destination.

Definition:

Switching is a networking technique in which data is transferred from a source to a destination through intermediate nodes that direct the data toward its destination.

Switching allows multiple devices to share network infrastructure without requiring a separate physical link between every pair of devices.

Importance of Switching

- It reduces the need for direct connections between every pair of devices.
- It allows communication between devices connected to different parts of a network.
- It improves the utilization of available communication links.
- It provides different methods for transferring data according to network requirements.

Types of Switching

There are three fundamental types of switching:

1. Circuit Switching
2. Message Switching
3. Packet Switching

Packet switching has two major approaches:

1. Virtual Circuit Switching
2. Datagram Switching

1 Circuit Switching

Definition:

Circuit switching is a switching technique in which a dedicated communication path is established between the source and destination before data transfer begins. The path and its required resources remain reserved for the duration of the connection.

Working Principle

Circuit switching works in three main phases:

1. **Connection Establishment:** The network establishes a dedicated path between the communicating devices. If the required resources are unavailable, the connection may be rejected.
2. **Data Transfer:** After the path is established, data travels through the reserved path.
3. **Connection Termination:** When communication ends, the connection is released and the reserved resources become available to other users.

Important Characteristics

- **Dedicated Path:** A dedicated communication path is established through the network.
- **Continuous Data Flow:** Once the connection is established, data can flow continuously without requiring each unit of data to be independently routed.
- **No Per-Packet Routing Headers:** Traditional circuit switching does not require each data unit to carry a complete destination address for independent routing because the path has already been established.
- **Lower Transfer Delay:** Data transfer can have predictable delay after setup because the path and resources are reserved.
- **Resource Utilization:** Reserved resources may remain idle when the communicating devices have no data to send, reducing efficiency.

Practical Example

Traditional telephone networks are a common example of circuit switching. When a telephone call is established, the network allocates a communication path for the call. The path remains reserved until the call ends.

Advantages

- Provides a dedicated path and predictable communication performance after setup.
- Supports continuous data transfer.
- Does not require independent routing decisions for each data unit.

Disadvantages

- Connection establishment introduces an initial delay.
- Reserved resources may be wasted when no data is being transferred.
- A connection may be blocked if the required resources are unavailable.

2 Message Switching

Definition:

Message switching is a switching technique in which the complete message is received and stored at an intermediate node before being forwarded to the next node toward its destination.

It follows the store-and-forward principle, but the unit stored and forwarded is the entire message rather than an individual packet.

Working Principle

1. The source sends the complete message to an intermediate switching node.
2. The node receives and stores the entire message in its memory.
3. The node forwards the message when the next communication link becomes available.
4. The process continues until the message reaches its destination.

Important Characteristics

- **No Dedicated Path:** A permanent, dedicated path is not required before sending the message.
- **Store and Forward:** Each intermediate node stores the complete message before forwarding it.
- **Storage Requirement:** Intermediate nodes need sufficient memory to store messages.
- **Variable Delay:** A message may wait in storage before the next link becomes available.

Practical Example

Consider a system that forwards a complete message from one computer to another through several intermediate computers. Each intermediate computer stores the complete message before forwarding it.

Message switching is mainly of historical and conceptual importance in networking education. Modern Internet communication generally uses packet switching rather than forwarding complete messages in this way.

Advantages

- Unlike circuit switching, it does not require a dedicated path to remain reserved for the entire communication.
- Communication links can be shared among different messages.
- Messages can wait in storage when the next link is temporarily unavailable.

Disadvantages

- Storing complete messages requires substantial memory, especially for large messages.
- The complete message must be received before it can be forwarded, increasing delay.
- It is less suitable for real-time voice and video communication where low delay is important.

3 Packet Switching

Definition:

Packet switching is a switching technique in which a message is divided into smaller units called packets. These packets are transferred through the network and reassembled at the destination when required.

Packet switching is the fundamental approach used for data communication across the Internet.

Working Principle

1. The source divides the message into smaller packets.
2. Each packet carries the information needed for delivery, including relevant addressing and control information.
3. Network devices receive and forward packets toward their destinations.
4. Packets are reassembled into the original message at the destination when required.

Important Characteristics

- **Data Link and Network Layers:** Packet forwarding can involve data-link switches and network-layer routers, depending on the network and device.
- **Store and Forward:** A packet switch commonly receives a complete packet before forwarding it onto the next link.
- **Pipelining:** Different packets can occupy different links at the same time. This allows transmission to overlap and can improve overall efficiency.
- **Efficient Resource Utilization:** Communication links are shared by packets from different users as needed.

- **Variable Delay:** Packets may experience queuing and processing delays, particularly when the network is busy.

Practical Example

When a user opens a website, the requested data is transferred through packets. Routers forward these packets through interconnected networks until they reach the destination. The receiving system processes the packets and reconstructs the required data.

Advantages

- Communication links are shared efficiently among many users.
- No dedicated end-to-end physical circuit is required for ordinary datagram communication.
- Pipelining allows multiple packets to be transmitted through different links concurrently.
- The network can route packets around some link or node failures when alternative routes are available.

Disadvantages

- Queuing and congestion can increase delay.
- Packets may be lost when network buffers become full.
- Packets may arrive out of order, depending on the switching approach and routes used.
- Packet headers introduce overhead because addressing and control information must be carried.

Types of Packet Switching

Packet switching is commonly divided into two approaches: virtual circuit switching and datagram switching.

A. Virtual Circuit Switching

Definition:

Virtual circuit switching is a connection-oriented packet-switching approach in which a logical path is established before data transfer. Packets belonging to the same virtual circuit generally follow the established logical path.

Working Principle

1. **Setup:** The network establishes a virtual circuit and determines the logical path.

2. **Data Transfer:** Packets are sent along the established path using virtual-circuit identifiers.
3. **Termination:** When communication ends, the virtual circuit and any associated reserved resources are released.

Important Characteristics

- **Connection-Oriented:** A virtual circuit is established before data transfer.
- **Logical Path:** Packets follow an established logical route rather than independently selecting a route at every node.
- **Same Order:** Packets generally follow the same path and are normally delivered in order, although failures can disrupt delivery.
- **Less Addressing Overhead:** Packets can use a short virtual-circuit identifier instead of carrying a complete destination address for independent routing at every switch.
- **Resource Reservation:** Some virtual-circuit systems can reserve resources or provide quality-of-service guarantees, but reservation is not a requirement of every virtual circuit.

Practical Examples

X.25 and ATM are classic examples of virtual-circuit technologies. They establish logical connections before transferring data.

Advantages

- Packets generally follow a consistent path and can arrive in order.
- Short virtual-circuit identifiers can reduce per-packet addressing overhead.
- Systems that support resource reservation can provide more predictable performance.

Disadvantages

- Connection setup introduces an initial delay.
- Network state must be maintained for established virtual circuits.
- A failure along the established path may interrupt communication and require recovery or circuit re-establishment.

B. Datagram Switching

Definition:

Datagram switching is a connectionless packet-switching approach in which each packet is routed independently using its destination address.

Important Characteristics

- **Connectionless:** A network-layer connection does not need to be established before sending packets.
- **No Resource Reservation:** A dedicated path or fixed set of resources is not reserved for the entire communication.
- **Independent Routing:** Different packets from the same message may follow different routes.
- **Out-of-Order Delivery:** Packets may arrive in a different order from the order in which they were sent.
- **Header Overhead:** Packets generally carry destination addressing information needed for independent routing.
- **Packet Loss:** Packets may be lost because of congestion, link failures, or full buffers.

Practical Example

The Internet Protocol (IP) uses datagram-based forwarding. When a computer sends data across the Internet, routers forward IP packets according to their destination IP addresses. Different packets may follow different routes.

Advantages

- No connection-establishment delay is required at the network layer.
- Network resources can be shared dynamically.
- Alternative routes may be used when network conditions change.

Disadvantages

- Packets may arrive out of order or fail to reach the destination.
- Each packet carries addressing information for independent forwarding.
- Delay and packet loss can vary with network conditions.

References

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